

Linear Algebra

Part-3

- 1- Create two vectors $a = [1, 2, 3]$ and $b = [4, 5, 6]$. Compute element-wise addition, subtraction, and dot product.
- 2- Multiply a 2×3 matrix $\begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \end{bmatrix}$ with a 3×2 matrix $\begin{bmatrix} 7 & 8 \\ 9 & 10 \\ 11 & 12 \end{bmatrix}$.
- 3- Transpose a matrix $\begin{bmatrix} 1 & 2 \\ 3 & 4 \\ 5 & 6 \end{bmatrix}$.
- 4- Create a 4×4 identity matrix.
- 5- Compute the inverse of $\begin{bmatrix} 4 & 7 \\ 2 & 6 \end{bmatrix}$.
- 6- Solve $Ax = b$ where $A = \begin{bmatrix} 3 & 1 \\ 1 & 2 \end{bmatrix}$ and $b = [9, 8]$.
- 7- Find eigenvalues and eigenvectors of $\begin{bmatrix} 2 & -1 \\ -1 & 2 \end{bmatrix}$.
- 8- Compute the determinant of $\begin{bmatrix} 1 & 2 \\ 3 & 4 \end{bmatrix}$.